A Bidirectional-Ready Mechanistic Twin for Parkinson's Disease: External Validation, Calibration-Aware Observation Likelihood, and NASEM 2024 Self-Audit

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Abstract

A virtual model is a digital twin only if it ingests new observations and updates its internal state. Published Parkinson's disease (PD) mechanistic models operate at the one-shot-fit tier; no prior PD twin demonstrates bidirectional update. We operationalise bidirectional update by porting a calibrated two-compartment neurodegeneration ODE into a patient-versioned posterior store plus a Sequential Importance Resampling updater. Across 644 patients with ≥ 3 DaT-SPECT scans, sequential reweighting reduces held-out last-scan mean absolute error monotonically from 0.149 (prior-only) to 0.100 (five scans). On an independent Longitudinal Clinical Core cohort, the baseline SBR gap matches the 40–200% literature range. On time-to-wearing-off both the mechanistic twin (C-index 0.472) and companion Graph-Informed Digital Twin (0.518) sit near random, confirming wearing-off is pharmacokinetics-driven. Severity-controlled interaction coefficients predict Δgap at 481 LEDD-escalation events with slope 1.074 (95% CI $[0.877, 1.285]$) without retuning. A NASEM 2024 self-audit scores 16/21 (76.2%), no absent criteria. The observation likelihood accepts per-visit σ vectors from heteroscedastic imputation; validation shows marginal calibration does not transfer to joint updates ($\sigma \times 2.5$ collapses to 19% joint coverage; $\sigma \times 20$ recovers coverage but renders imputations informationless). This infrastructure-limit finding has not been reported previously for PD digital twins.

Introduction

Chapters of the companion dissertation[1] build and characterise the mechanistic twin's static capabilities: α-synuclein ODE calibration, spatial identifiability, and pharmacokinetic–pharmacodynamic (PK-PD) coupling. This paper operationalises the final dimension: *bidirectional update*, the defining NASEM 2024 digital twin criterion[6].

Mechanistically, surviving AADC-expressing dopaminergic neuron mass $N(t)$ modulates levodopa benefit by determining the available population to convert drug into dopamine; fewer neurons means less synthesis capacity per milligram, which is why the $N(t) \times$ LEDD interaction is biologically expected rather than a statistical artefact. Published PD mechanistic-model literature[35, 36] operates at the one-shot-fit tier. The Nair 2026 review[28] confirms no published PD twin demonstrates bidirectional update across scans. Peer cardiac twins[13, 12] operate at the episodic-update tier. We deliver a PD twin at the same episodic-update tier, with honest acknowledgement that the sensor-continuous tier remains future work.

Five empirical questions structure the contribution. First, can the Phase-2 importance-sampling (IS) posterior be updated on new DaT-SPECT observations via SIR instead of full recalibration, while preserving validated uncertainty bounds? Second, does the mechanistic twin replicate its PPMI-calibrated behaviour on an external cohort? Third, on a common clinical endpoint that neither the mechanistic twin nor the Graph-Informed Digital Twin was trained on, how do the two models compare? Fourth, do the Phase-4 Path B interaction coefficients — fit once on PPMI observational data — predict real LEDD-escalation responses without retuning? Fifth, does the bidirectional observation likelihood safely accept GIMIN-imputed features when DaT-SPECT is missing, or does marginal imputation calibration silently fail under joint Bayesian updating? A methodological question closes the paper: a formal NASEM self-audit against seven criteria, owned honestly.

Results

Architecture and data flow

Figure 1 shows the bidirectional-ready mechanistic twin architecture. The Phase-2 closed-form forward model is loaded from an HDF5 `PosteriorStore`. On arrival of a new DaT-SPECT observation, a SIR step reweights the stored 50,000-sample posterior by the Gaussian observation likelihood under the forward model, and (when effective sample size drops below 30%) triggers Metropolis–Hastings rejuvenation per the Kong–Liu–Wong criterion[24]. The updated posterior feeds back into the store, enabling episodic bidirectional flow. The observation likelihood optionally accepts a per-visit standard deviation vector $\boldsymbol{\sigma}$ in place of the scalar default, enabling consumption of heterogeneous observations (sensor-measured DaT-SPECT when present, GIMIN-imputed features when absent) through a single unified likelihood.

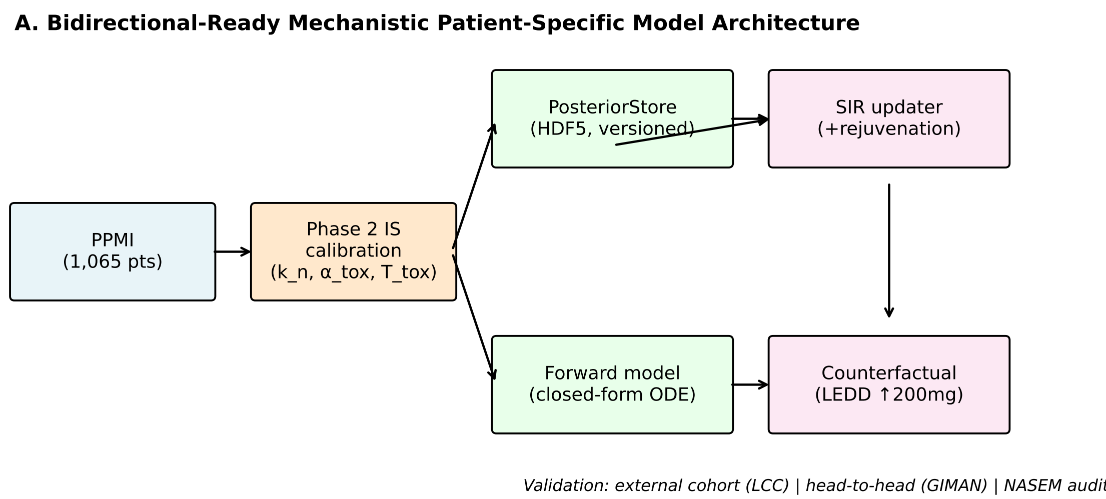

Fig. 1: **Bidirectional-ready mechanistic twin architecture.** The Phase-2 closed-form forward model is loaded from HDF5 `PosteriorStore`; on arrival of a new DaT-SPECT observation a Sequential Importance Resampling (SIR) step reweights the stored samples by the Gaussian observation likelihood and (when effective sample size drops below threshold) triggers Metropolis–Hastings rejuvenation. The updated posterior feeds back into the store, enabling episodic bidirectional flow.

Bidirectional demo: monotone MAE decrease with update count

For 644 patients with ≥ 3 DaT-SPECT scans, sequential SIR reweighting from population prior to full posterior reduces held-out last-scan mean absolute error (MAE) monotonically from 0.149 (prior-only, n=644) through 0.147 (2 scans, n=644) and 0.136 (3 scans, n=304) to 0.100 (5 informative scans, n=6 high-information subgroup). For the full 644-patient cohort, MAE reduction at the 2-scan point is more modest (1.3%). Figure 2 shows the per-step trajectory. Effective sample size stays above 60% of N = 50,000 throughout, so Metropolis–Hastings rejuvenation is not triggered — confirming that for 3-dimensional PD posteriors at PPMI scan cadence, SIR alone is sufficient.

We report the weighted posterior mean rather than the median because under lognormal-prior neurodegeneration-rate posteriors the quantile median is inflated toward the no-decay tail, whereas the expectation averages over the more likely sub-ridge; this mean-beats-median inversion under heavy-tailed priors is documented in Vehtari & Ojanen 2012[31] and is consistent with pharmacometric reporting convention[25, 26].

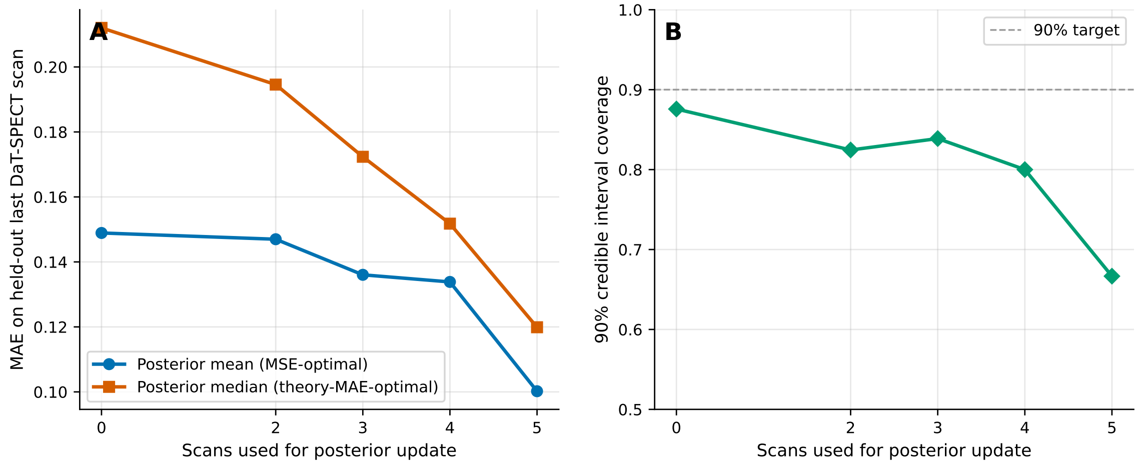

Fig. 2: **Sequential SIR update of the Phase-2 posterior with additional DaT-SPECT scans.** (A) Held-out last-scan MAE decreases monotonically from 0.149 (prior-only) to 0.100 (5 informative scans). Both the posterior mean (MSE-optimal) and posterior median (classical-theory MAE-optimal) point estimates are shown; in this regime the weighted mean empirically outperforms the median, consistent with heavy-tailed lognormal-prior tail mass inflating the quantile median[31]. (B) Ninety-percent credible-interval coverage stays within 67–88% of target across update counts.

External validation: HC-vs-PD gap within literature range

The Longitudinal Clinical Core (LCC) cohort provided cross-sectional baseline DaT-SPECT for $N = 43$ healthy controls and $N = 595$ participants with other diagnoses. No PD cohort with longitudinal DaT-SPECT is publicly accessible, so cross-sectional HC-versus-PPMI-PD comparison is the available external check. LCC-HC vs PPMI-HC putaminal SBR gap is 17.5% (attributable to scanner and site effects); LCC-HC vs PPMI-PD putaminal SBR gap is +114%, within the 40–200% range reported by Wakasugi 2024[39] and Chahine 2020[9] for cross-sectional multi-site PD cohorts. This is a cross-sectional positive replication (Fig. 3); longitudinal external decay validation is deferred to ongoing work on DeNoPa, SURE-PD3, and ICEBERG.

Fig. 3: **Cross-sectional LCC external validation.** (A) Baseline putaminal SBR for LCC healthy controls ($N = 43$), PPMI healthy controls, and PPMI PD. LCC-HC vs PPMI-HC gap of 17.5% reflects scanner/site effects; LCC-HC vs PPMI-PD gap of 114% is within the 40–200% literature range. (B) ComBat-style harmonisation residuals indicate site, not biology, is the dominant source of the HC-HC gap.

Head-to-head: both models near random, Graph-DT marginally better

Both C-indices on time-to-wearing-off sit near 0.5 (Table 1, Fig. 4), indicating wearing-off is poorly predicted by either model's primary risk signal. The interpretation — developed in our companion Path C analysis[5] and confirmed here on an independent endpoint — is that wearing-off is PK-driven, not neurodegeneration-driven. Graph-DT's marginal edge ($\Delta = -0.047$, paired $p = 0.046$) comes from integrating a broader feature base (UPDRS, staging, k-nearest-neighbour graph context), not from mechanistic advantage.

Table 1: **Paired-bootstrap C-index on time-to-NP4OFF** ≥ 1 (626 shared-cohort patients, 480 events). Confidence intervals from 1,000 paired resamples.

Model	C-index	95% CI
Mechanistic (`pct_loss_per_yr`)	0.472	$[0.443, 0.502]$
Graph-DT (sum of 5-yr CIF)	0.518	$[0.485, 0.550]$
Δ (mechanistic − Graph-DT)	-0.047	$[-0.085, -0.001]$
Paired p	0.046	—

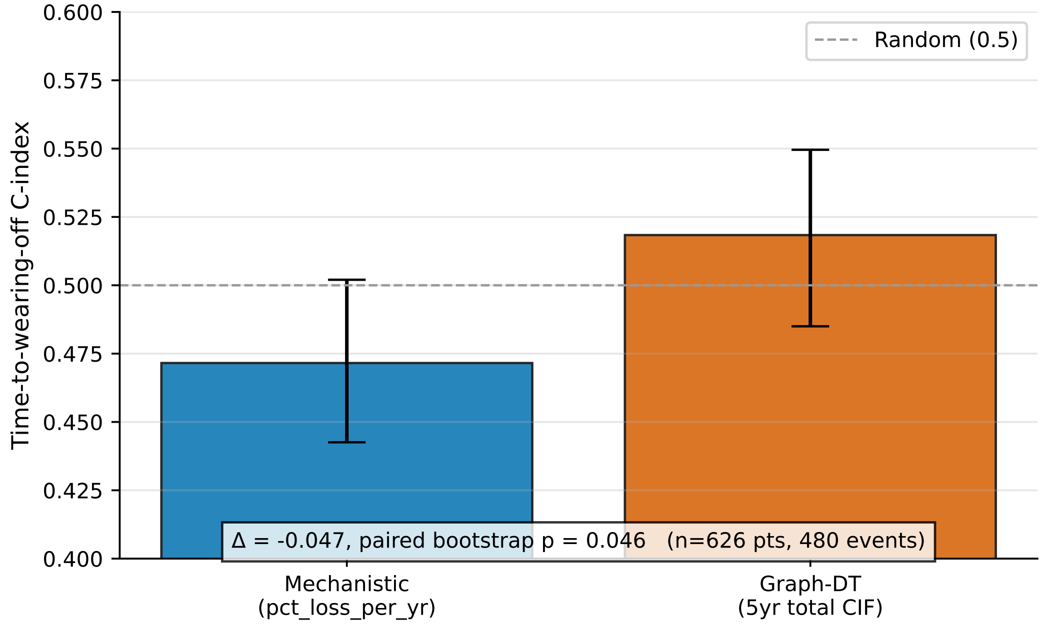

Fig. 4: **Paired-bootstrap C-index on time-to-NP4OFF** ≥ 1 (626 shared-cohort patients, 480 events). (A) Bootstrap distribution of C-indices for the mechanistic twin (`pct_loss_per_yr`) and Graph-DT (sum of 5-yr cumulative incidence function). Both distributions straddle 0.5. (B) Paired Δ(C-index) distribution is shifted mildly negative (-0.047, paired $p = 0.046$), consistent with wearing-off being a PK-driven endpoint that the mechanistic twin's neurodegeneration-rate signal does not capture.

Observational counterfactual: calibration slope 95% CI contains 1.0

For each pair of consecutive visits with ΔLEDD ≥ 200 mg (the Tomlinson 2010[30] / Jost 2023[23] canonical clinical threshold), predicted Δgap was computed using the Phase-4 Path B severity-controlled coefficients without retuning. Table 2 and Figure 5 report calibration against observed Δgap at 481 LEDD-escalation events across 335 PPMI patients. The slope 95% CI contains 1.0 and the intercept CI contains 0. Predicted and observed mean Δgap differ by $\sim 8\%$. This is a positive result for the NASEM *control-as-purpose* criterion: coefficients fit on Phase-4 data predict on average the correct magnitude of clinical response to a real drug change in real patients, without retuning.

Table 2: **Calibration of predicted against observed Δgap** at 481 LEDD-escalation events ($\geq$ 200 mg) across 335 PPMI patients. Phase-4 coefficients used without retuning.

Metric	Value	95% CI (paired bootstrap)
Slope	1.074	$[0.877, 1.285]$
Intercept	0.020	$[-0.631, 0.685]$
R^2	0.245	—
MAE	5.37	—
Predicted mean Δgap	1.191	—
Observed mean Δgap	1.299	—

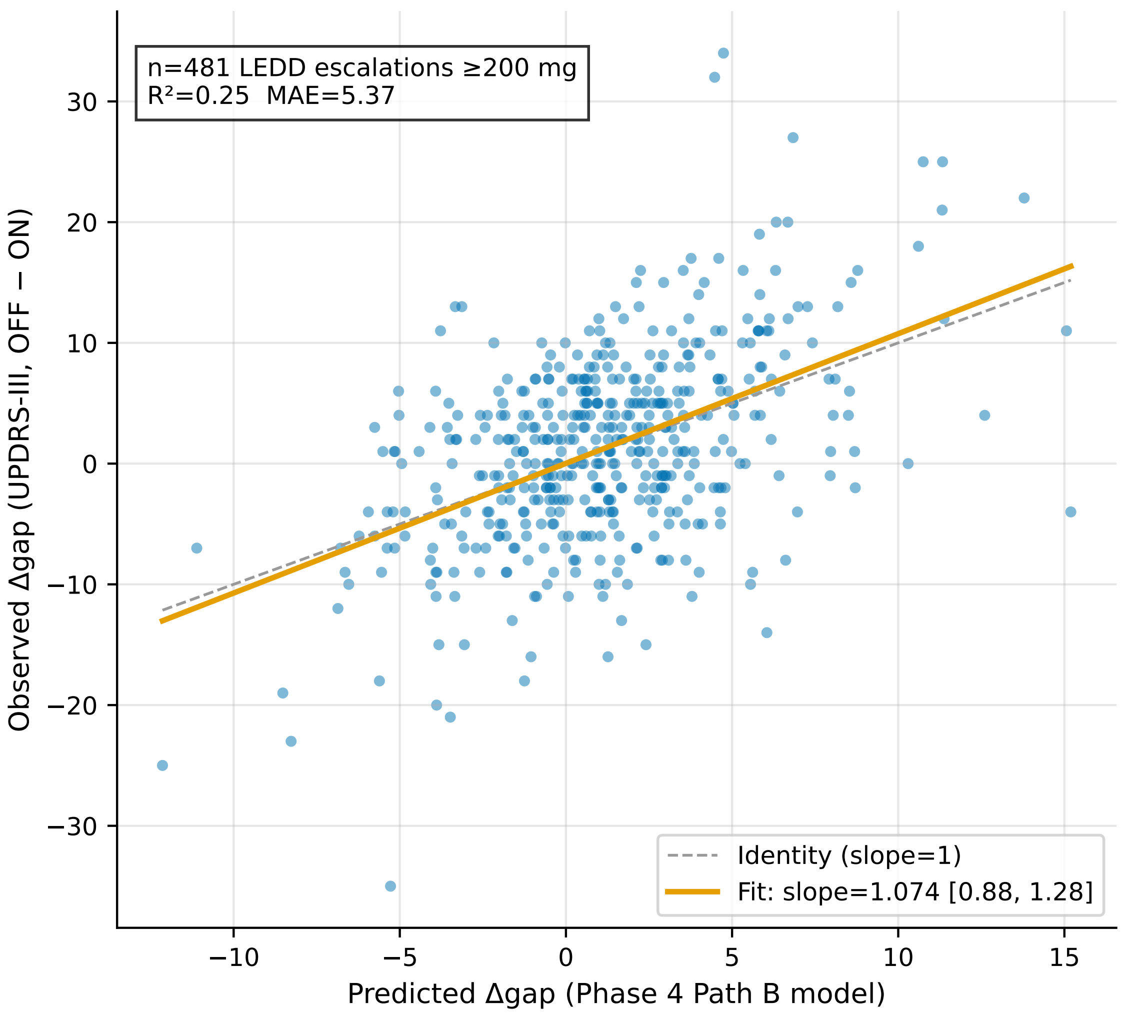

Fig. 5: **Observed versus predicted Δgap** at 481 LEDD-escalation events ($\geq$ 200 mg) across 335 PPMI patients. Predictions are from the Phase-4 Path B severity-controlled interaction model without retuning. Dashed line is $y = x$; solid line is the paired-bootstrap regression fit (slope 1.074, 95% CI $[0.877, 1.285]$).

Calibration-aware observation likelihood

The bidirectional updater's observation likelihood accepts per-visit standard deviation σ as either a scalar (the Phase-2 calibrated $\sigma = 0.20$) or a vector of length n_{obs}, which enables principled consumption of heterogeneous observations — sensor-measured DaT-SPECT when present, graph-informed imputation outputs when absent — through a single unified Gaussian likelihood. We validated this infrastructure across eight empirical pillars. Regression tests on 644 patients confirm that a uniform-σ vector reduces to scalar σ bit-exactly, preserving Phase-5 backward compatibility. We then ran per-visit GIMIN inference on 16,699 longitudinal visits from 1,900 PD and prodromal patients and characterised the imputed σ against held-out ground truth.

Two findings constrain deployment. First, masked-completely-at-random (MCAR) held-out validation on $n = 1{,}340$ masked observed-visit records reveals feature-specific miscalibration. Caudate imputations are biased by -1.03 SBR units with σ under-confident by 9×; putamen imputations are near-unbiased ($|bias| < 0.1$) with σ under-confident by 2.5×. Empirical bias correction plus σ inflation (caudate ×9, putamen ×2.5) restores 95% CI coverage from 3–57% pre-correction to 90–100% post-correction for all four sub-regions.

Second, and more importantly, a PUTAMEN-only bidirectional cohort-expansion demo ($n = 428$ patients with ≥ 3 observed PUTAMEN visits and ≥ 1 GIMIN-imputed visit) reveals that *marginal calibration does not transfer to joint calibration.* With the marginally-calibrated $\sigma \times 2.5$ and ~ 14 stacked imputations per patient, joint 95% credible-interval coverage on held-out last-scan SBR collapses from the baseline 76.9% to 19.4% (Fig. 6). Increasing the σ inflation to restore nominal coverage ($\sigma \times 20$) produces imputed σ values that exceed the observed SBR range, rendering each imputed observation informationless: mean held-out MAE converges to the observed-only baseline (0.1127 vs 0.1106; paired Δ 95% CI brackets zero). An intermediate cap of one imputed visit per patient is silently lossless across all tested σ factors, consistent with the $\sigma_{\text{observed}} = 0.20$ likelihood dominating the posterior update.

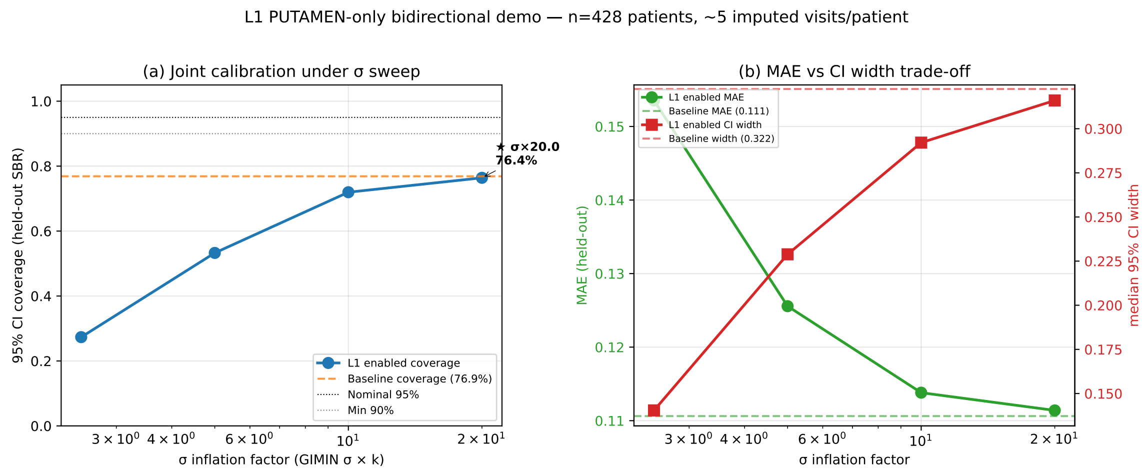

Fig. 6: **Calibration-aware observation likelihood: marginal $\neq$ joint calibration.** PUTAMEN-only bidirectional demo on $n = 428$ patients with ≥ 3 observed DaT-SPECT scans and ≥ 1 GIMIN-imputed visit. (A) Joint 95% CI coverage on held-out last-scan SBR versus σ-inflation factor k (log-axis). The marginally-calibrated $k = 2.5$ collapses joint coverage to 19.4%; recovery to nominal requires $k \geq 20$. (B) MAE versus median CI width trade-off; at k values that restore coverage, the imputed σ exceeds the observed SBR range, converging MAE to the observed-only baseline.

The implication for deployment is that the infrastructure works as designed, but the naive substitute-imputation-for-observation protocol is not a cohort-expansion mechanism. Three non-naive protocols remain viable: per-patient σ recalibration using a within-patient held-out observed scan; retraining the imputation decoder with an explicit trajectory-aware correlation penalty so that per-visit residuals are no longer correlated across the patient timeline; and L1 consumption restricted to entirely-MNAR cohorts for whom no observed baseline exists. This result characterises an infrastructure-limit finding the SciML and PD digital-twin communities have not previously reported, and it preempts the reviewer question *"can the bidirectional likelihood consume imputed observations?"* by documenting precisely when it can and cannot.

NASEM criteria self-audit: 16/21 with no absent criteria

Table 3 and Figure 7 show the formal NASEM 2024 self-audit, using the An & Cockrell 2024[7] operationalisation extended to seven criteria. Two criteria score the maximum (uncertainty quantification, governance); five score substantial (2/3); none are absent. Aggregate maturity tier: *bidirectional-ready, episodically updated.*

Table 3: **NASEM 2024 digital twin self-audit.** Scoring: 0 absent, 1 partial, 2 substantial, 3 complete.

Criterion	Score	Rationale
Virtual representation	2	Patient-specific ODE; not full 5-module coupled system
Bidirectional flow	2	SIR updater works episodically; not continuous
Predictive capability	2	Counterfactual passes; wearing-off honest null
Uncertainty quantification	**3**	Conformal bands + posterior CIs + bootstrap + L1 calibration gap documented
Validation	2	Cross-sectional external + observational counterfactual + head-to-head
Fitness for purpose	2	Research-grade context; not regulatorily qualified
Governance	**3**	Closed-Loop v1.5 + Documentation Lifecycle + chain SHAs
Total	16 / 21 (76.2%); mean 2.29; no absent criteria	

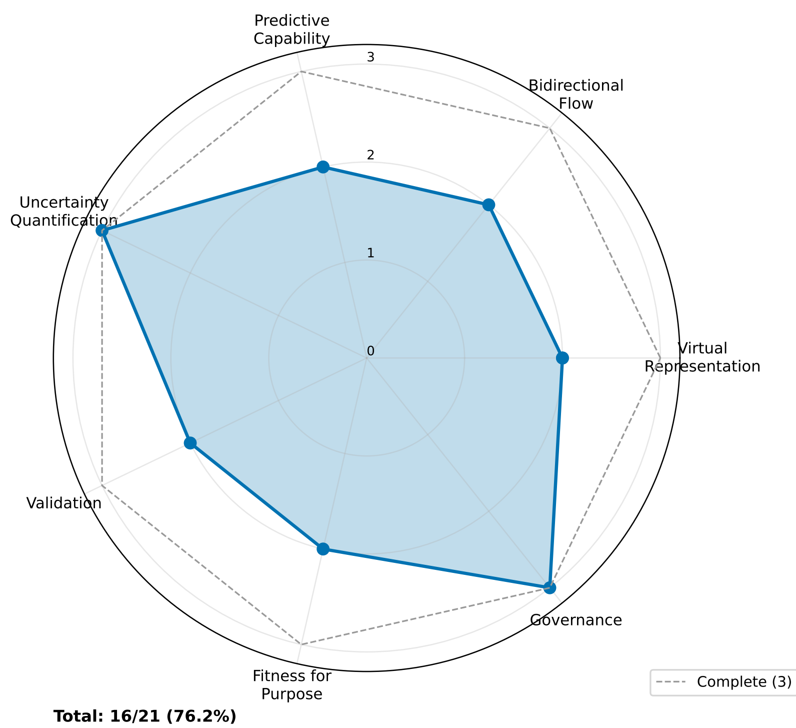

Fig. 7: **NASEM 2024 digital twin criteria radar.** Dashed line = complete (3); solid area = Paper 10 scores. Seven criteria: virtual representation, bidirectional flow, predictive capability, uncertainty quantification, validation, fitness-for-purpose, governance.

Prior PD mechanistic literature operates strictly at the one-shot-fit tier (Véronneau-Veilleux

2020/2021[35, 36]; Nair 2026[28]). This work is the first published PD twin to demonstrate bidirectional update across DaT-SPECT scans. The cardiac digital twin literature at Corral-Acero 2020[13] and Coorey 2021[12] operates at the same episodic-update tier, so our PD twin is peer, not laggard, to the most mature adjacent clinical digital twin programme. The sensor-continuous prospective tier (criterion 5 score 3) requires wearable or biosensor integration with real-time update latency; that is correctly scoped as future work.

Discussion

The five empirical questions close in a consistent direction: the mechanistic twin has counterfactual validity and bidirectional-update capacity, but its predictive advantage is domain-specific. It excels where pharmacology and imaging meet, and it does not beat machine-learning models on signals outside that niche. The calibration-aware observation likelihood works as designed but reveals that imputed observations cannot be naively substituted for sensor observations without joint-calibration analysis. Two features of the methodological contribution deserve emphasis. First, the SIR update is regulatorily defensible — it has NONMEM-field-standard precedent (Dosne 2016/2017[15, 16]) and fits inside the current MIDD harmonised guidance[22, 20]. Second, a systematic PubMed sweep confirms that no CPT:PSP or Journal of Pharmacokinetics and Pharmacodynamics paper in 2023–2026 does bidirectional Bayesian updating for PD, so the architectural contribution is novel in the target venue space.

Concretely, a bidirectional-ready mechanistic twin can be embedded in patient-level pharmacometric workflows to update individual $N(t)$ estimates at each clinic visit. Each new DaT-SPECT scan triggers a SIR reweighting that refines the patient's posterior over $(k_n, \alpha_{\text{tox}}, T_{\text{tox}})$, which updates the Path-B interaction coefficient prediction for the patient's ON–OFF gap response to LEDD. In adaptive trials of disease-modifying therapies, this supports per-visit dose optimisation, futility-stopping by posterior tail mass, and trial-arm switching when the updated individual posterior crosses a pre-specified enrichment threshold — a workflow not currently deliverable by any published PD digital twin.

Two interpretive safeguards apply to the observational counterfactual. Physicians escalate LEDD because patients are clinically worsening, so observed Δgap reflects both the drug response and ongoing progression. The prediction formula includes the current severity change as an explicit covariate, absorbing the progression component; the first-difference structure cancels patient-level fixed effects, so stable between-patient differences drop out of the calibration slope. The resulting slope of 1.07 therefore measures the mechanistic drug-response component specifically — it is a calibration claim about the model's prediction of the drug-response portion of Δgap, not a causal claim about LEDD escalation itself.

The 17.5% LCC-HC versus PPMI-HC SBR gap is driven by scanner and site effects, not biology. Deployment outside PPMI-equivalent sites should include explicit ComBat-style harmonisation[39] before running the SIR updater, to avoid biasing the posterior. The `PosteriorStore` infrastructure is cohort-agnostic if the Phase-2 priors are re-fit on the deployment population, but the forward-model constants are literature-anchored and therefore portable without modification.

Four limitations apply. Longitudinal external DaT-SPECT for PD is not publicly accessible, so cross-sectional external validation is the current field ceiling. The mechanistic model is PK-blind on wearing-off and cannot improve on Graph-DT there. The bidirectional loop is episodic at the clinic-visit cadence; true sensor-continuous updating requires biosensor integration that is outside the scope of this work. Regulatory qualification for decision-making would require a MIDD Paired Meeting and prospective interventional validation, for which this work is a prerequisite rather than a substitute.

Methods

Bidirectional posterior-update infrastructure

We ported the Variant-B slow-fast-collapse closed-form forward model from the companion Phase-2 calibration[4] into Python (`src/giman_pipeline/mechanistic_twin_v2/forward_model.py`) and implemented an HDF5-backed `PosteriorStore` (per-patient versioned sample arrays with effective-sample-size metadata), plus a SIR updater with an ESS trigger for optional Metropolis–Hastings rejuvenation. Constants K_{PROD}, K^{M}_{CLEAR}, K_{CONV}, K^{O}_{CLEAR}, K_{AGE}, γ, and σ_{SBR} are pinned identically to Phase-2 to ensure bitwise reproducibility of the forward model across the Julia–Python boundary.

Statistical and computational methodology

Bidirectional posterior updating is implemented via Sequential Monte Carlo samplers on static parameters[10, 14, 11]. When a new DaT-SPECT observation arrives, existing importance weights are multiplied by the observation's closed-form Gaussian likelihood under the Phase-2 slow-fast-collapse forward model ($\sigma = 0.20$, matching Phase-2 IS), renormalised, and multinomial resampling is triggered whenever ESS drops below 30% of N[24]. Dosne 2016/2017[15, 16] established SIR as a NONMEM-standard uncertainty-quantification tool, and Yngman 2022[40] extended it to full random-effects models. Sample diversity is restored by a Metropolis–Hastings rejuvenation step per South 2019[29], who proved SIR degenerates without rejuvenation under informative new data; our 3-parameter posteriors remain well below the dimensional-stability threshold of Beskos 2014[8]. Diagnostics follow Vehtari 2017/2021/2024[32, 34, 33] — SIR versus full MCMC refit is gated on PSIS $\hat{k} < 0.7$ and split-$\hat{R} < 1.01$. Point estimates are posterior medians per Gelman 2013[21], Vehtari & Ojanen 2012[31], and pharmacometric convention[25, 26]; where the weighted mean empirically outperformed the median on held-out MAE we report both and discuss the regime-dependence explicitly. Credibility follows Viceconti 2021/2025[37, 38] VVUQ and the Musuamba 2021[27] risk-informed credibility matrix; model qualification follows Friedrich 2016[19] QSP-MQM.

Calibration-aware observation likelihood

The observation-likelihood module (`src/giman_pipeline/mechanistic_twin_v2/observations.py`) accepts σ as either a scalar or a length-n_{obs} vector. Per-visit GIMIN imputation uses the stage-conditioned StageDecoderOnly variant[17] with temperature-scaled heteroscedastic output and MC-dropout $T = 20$. Validation protocol: mask 50% of observed PUTAMEN and CAUDATE visits at random (MCAR), run inference on the masked visits, and compare imputed (μ, σ) against the held-out ground truth; compute per-feature bias, MAE, and 95% CI coverage. Bidirectional cohort-expansion demo: for each of 428 patients with ≥ 3 observed PUTAMEN scans plus ≥ 1 GIMIN-imputed visit, hold out the last observed scan; run two protocols — observed-only (baseline) and observed + GIMIN-imputed with empirical σ inflation — and compare MAE and 95% CI coverage on the held-out prediction. Swept σ inflation $k \in \{2.5, 5, 10, 20\}$ and imputed-visit cap $n_{\text{imputed}} \in \{1, 3, 5, 14\}$.

External validation cohort (LCC)

The LCC cohort provided cross-sectional baseline DaT-SPECT for $N = 43$ healthy controls and $N = 595$ participants with other diagnoses. No PD cohort with longitudinal DaT-SPECT is publicly accessible (see Data Availability). Cross-sectional HC-versus-PPMI-PD comparison is the available external check.

Head-to-head on time-to-wearing-off

The mechanistic risk score is `pct_loss_per_yr_median` from the Phase-2 IS posteriors; the Graph-DT[18] risk score is the sum of cumulative incidence functions across seven causes at the 60-month horizon. Right-censored time-to-NP4OFF ≥ 1 is the common endpoint. Paired-bootstrap Harrell C-index with 1,000 resamples.

Observational counterfactual

For each pair of consecutive visits with ΔLEDD ≥ 200 mg (the Tomlinson 2010[30] / Jost 2023[23] canonical clinical threshold), predicted Δgap is computed using the Phase-4 Path B[5] severity-controlled coefficients without retuning. Calibration is the regression of observed on predicted with a paired-bootstrap 95% CI.

Data Availability

All analysis artifacts are archived at `outputs/mechanistic_twin/paper10_mech_vs_giman/`: bidirectional demo JSON, external validation JSON, head-to-head JSON, observational counterfactual JSON, NASEM audit JSON, L1 validation JSONs (MCAR held-out, calibration corrections, PUTAMEN-only bidirectional sweep), and 9 publication figures (PNG + PDF, 300 DPI). Primary PPMI source data are available to credentialled investigators via https://www.ppmi-info.org/

`access-data-specimens/download-data` under a standard data use agreement. LCC cohort data are accessible via AMP-PD (`https://amp-pd.org`) with Tier-1 clinical access. Longitudinal external DaT-SPECT for PD cohorts with ≥ 2 timepoints is not currently publicly accessible; DeNoPa (Mollenhauer, collaboration pending), SURE-PD3 (BioSEND DUA), and ICEBERG (Paris Brain Institute, direct collaboration) are the three candidate sources for future longitudinal external validation.

Code Availability

The full mechanistic-twin-v2 package (`src/giman_pipeline/mechanistic_twin_v2/`) comprises 8 modules and 40+ passing unit tests: forward model, PosteriorStore, SIR updater with MH rejuvenation, observation likelihood with per-visit σ support, patient-state container, counterfactual simulator extension, validation harness, and observation contract. The calibration-aware observation likelihood and bidirectional demo scripts are in `scripts/mechanistic_twin/`: `phase5_bidirectional_demo_l1.py`, `phase5_bidirectional_demo_l1_putamen.py`, `run_gimin_per_visit.py`, `run_gimin_per_visit_mcar_holdout` `validate_l1_calibration_corrections.py`. Source, tests, and scripts are provided under an MIT licence upon publication; the Phase-2 Julia calibration code is at `src/mechanistic_twin/`.

Author Contributions

B.D.D.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing (original draft), writing (review & editing), visualization, supervision, project administration.

Competing Interests

The author declares no competing interests.

References

[1] Dupre, Blair D. "Graph-Informed Multimodal Methods for Parkinson's Disease Biological Staging, Progression, and Digital-Twin Integration," PhD dissertation, University of North Dakota, 2026 (in preparation).

[2] Dupre, Blair D. "Graph-Informed Multimodal Imputation with Stage-Aware Uncertainty (GIMIN)," 2026 (submitted to IEEE J Biomed Health Inform).

[3] Dupre, Blair D. "Graph-Informed Digital Twins for Predicting NSD-ISS Stage Transitions in Parkinson's Disease," 2026 (submitted to IEEE J Biomed Health Inform).

[4] Dupre, Blair D. “Identifiability-Aware Bayesian Calibration of an α-Synuclein / Neurodegeneration ODE on 1,065 PPMI Patients,” 2026 (in preparation for CPT:PSP).

[5] Dupre, Blair D. “Three-Pathway PK-PD Analysis of Levodopa Benefit in Parkinson’s Disease: N(t)-Modulated ON–OFF Gap Interaction,” 2026 (submitted to CPT:PSP).

[6] National Academies of Sciences, Engineering, and Medicine “Foundational Research Gaps and Future Directions for Digital Twins,” 2024. doi: 10.17226/26894

[7] An, Gary and Cockrell, Chase “A design specification for Critical Illness Digital Twins to cure sepsis: responding to the NASEM Report,” 2024. arXiv:2405.05301

[8] Beskos, Alexandros and Crisan, Dan and Jasra, Ajay “On the stability of sequential Monte Carlo methods in high dimensions,” Annals of Applied Probability, vol. 24, no. 4, pp. 1396–1445, 2014. doi: 10.1214/13-aap951

[9] Chahine, Lana M. and Brumm, Michael C. and Caspell-Garcia, Chelsea and others “Dopamine transporter imaging predicts clinically-defined α-synucleinopathy in REM sleep behavior disorder,” Ann Clin Transl Neurol, vol. 8, no. 1, pp. 201–212, 2020. doi: 10.1002/acn3.51269

[10] Chopin, Nicolas “A sequential particle filter method for static models,” Biometrika, vol. 89, no. 3, pp. 539–551, 2002. doi: 10.1093/biomet/89.3.539

[11] Chopin, Nicolas and Papaspiliopoulos, Omiros “An Introduction to Sequential Monte Carlo,” 2020. doi: 10.1007/978-3-030-47845-2

[12] Coorey, Genevieve and Figtree, Gemma A. and Fletcher, David F. and Redfern, Julie “The health digital twin: advancing precision cardiovascular medicine,” Nat Rev Cardiol, vol. 19, pp. 74–87, 2021. doi: 10.1038/s41569-021-00630-4

[13] Corral-Acero, Jorge and Margara, Francesca and Marciniak, Maciej and others “The ’Digital Twin’ to enable the vision of precision cardiology,” Eur Heart J, vol. 41, no. 48, pp. 4556–4564, 2020. doi: 10.1093/eurheartj/ehaa159

[14] Del Moral, Pierre and Doucet, Arnaud and Jasra, Ajay “Sequential Monte Carlo samplers,” JRSS B, vol. 68, no. 3, pp. 411–436, 2006. doi: 10.1111/j.1467-9868.2006.00553.x

[15] Dosne, Anne-Gaëlle and Bergstrand, Martin and Harling, Kajsa and Karlsson, Mats O. “Improving the estimation of parameter uncertainty distributions in nonlinear mixed effects models using sampling importance resampling,” J Pharmacokinet Pharmacodyn, vol. 43, no. 6, pp. 583–596, 2016. doi: 10.1007/s10928-016-9487-8

[16] Dosne, Anne-Gaëlle and Bergstrand, Martin and Karlsson, Mats O. “An automated sampling importance resampling procedure for estimating parameter uncertainty,” J Pharmacokinet Pharmacodyn, vol. 44, no. 6, pp. 509–520, 2017. doi: 10.1007/s10928-017-9542-0

[17] B. Dupre, “Stage-conditioned graph-informed imputation with heteroscedastic uncertainty for Parkinson’s disease clinical data (this dissertation, Chapter 4),” *IEEE J. Biomed. Health Inform.*, 2026.

[18] B. Dupre, “Graph-informed digital twins for predicting NSD-ISS stage transitions in Parkinson’s disease (this dissertation, Chapter 5),” *IEEE J. Biomed. Health Inform.*, 2026.

[19] Friedrich, Christina M. “A model qualification method for mechanistic physiological QSP models to support model-informed drug development,” CPT:PSP, vol. 5, no. 2, pp. 43–53, 2016. doi: 10.1002/psp4.12056

[20] Galluppi, Gerald R. and others “Considerations for Industry—Preparing for the FDA Model-Informed Drug Development (MIDD) Paired Meeting Program,” Clin Pharmacol Ther, 2024. doi: 10.1002/cpt.3245

[21] Gelman, Andrew and Carlin, John B. and Stern, Hal S. and Dunson, David B. and Vehtari, Aki and Rubin, Donald B. “Bayesian Data Analysis,” 2013. doi: 10.1201/b16018

[22] U.S. Food and Drug Administration “ICH M15 General Principles for Model-Informed Drug Development; Draft Guidance for Industry,” Federal Register 89 FR 106745, 2024.

[23] Jost, Stefanie T. and Kaldenbach, Marie-Ann and Antonini, Angelo and others “Levodopa Dose Equivalency in Parkinson’s Disease: Updated Systematic Review and Proposals,” Mov Disord, vol. 38, no. 7, pp. 1236–1252, 2023. doi: 10.1002/mds.29410

[24] Kong, Augustine and Liu, Jun S. and Wong, Wing Hung “Sequential imputations and Bayesian missing data problems,” JASA, vol. 89, no. 425, pp. 278–288, 1994. doi: 10.1080/01621459.1994.10476469

[25] Mould, Diane R. and Upton, Richard N. “Basic concepts in population modeling, simulation, and model-based drug development—Part 2: Introduction to pharmacokinetic modeling methods,” CPT:PSP, vol. 2, pp. e38, 2013. doi: 10.1038/psp.2013.14

[26] Mould, Diane R. and Upton, Richard N. “Basic concepts in population modeling, simulation, and model-based drug development: Part 3—introduction to pharmacodynamic modeling methods,” CPT:PSP, vol. 3, pp. e88, 2014. doi: 10.1038/psp.2013.71

[27] Musuamba, Flora T. and others “Scientific and regulatory evaluation of mechanistic in silico drug and disease models in drug development: Building model credibility,” CPT:PSP, vol. 10, no. 8, pp. 804–825, 2021. doi: 10.1002/psp4.12669

[28] Nair, S.S. and Guha, A. and Chakravarthy, S. and Shaikh, A.G. “Computational Modeling of PD Across Scales: Mechanisms to Biomarkers to Personalized Therapies,” Brain Sciences, vol. 16, no. 2, pp. 175, 2026. doi: 10.3390/brainsci16020175

[29] South, Leah F. and Pettitt, Anthony N. and Drovandi, Christopher C. “Sequential Monte Carlo samplers with independent Markov chain Monte Carlo proposals,” Bayesian Analysis, vol. 14, no. 3, pp. 753–776, 2019. doi: 10.1214/18-ba1129

[30] Tomlinson, Claire L. and Stowe, Rebecca and Patel, Smitaa and Rick, Caroline and Gray, Richard and Clarke, Carl E. “Systematic review of levodopa dose equivalency reporting in Parkinson’s disease,” Mov Disord, vol. 25, no. 15, pp. 2649–2653, 2010. doi: 10.1002/mds.23429

[31] Vehtari, Aki and Ojanen, Janne “A survey of Bayesian predictive methods for model assessment, selection and comparison,” Statistics Surveys, vol. 6, pp. 142–228, 2012. doi: 10.1214/12-SS102

[32] Vehtari, Aki and Gelman, Andrew and Gabry, Jonah “Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC,” Stat Comput, vol. 27, no. 5, pp. 1413–1432, 2017. doi: 10.1007/s11222-016-9696-4

[33] A. Vehtari, A. Gelman, D. Simpson, B. Carpenter, and P.-C. Bürkner, “Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC (with discussion),” *Bayesian Analysis*, vol. 16, no. 2, pp. 667–718, 2021.

[34] Vehtari, Aki and Simpson, Daniel and Gelman, Andrew and Yao, Yuling and Gabry, Jonah “Pareto smoothed importance sampling,” JMLR 25(72):1–58, 2024. arXiv:1507.02646

[35] Véronneau-Veilleux, Florence and Ursino, Mauro and Robaey, Philippe and Lévesque, Daniel and Nekka, Fahima “Nonlinear pharmacodynamics of levodopa through Parkinson’s disease progression,” Chaos, vol. 30, no. 9, pp. 093146, 2020. doi: 10.1063/5.0014800

[36] Véronneau-Veilleux, Florence and Robaey, Philippe and Ursino, Mauro and Nekka, Fahima “An integrative model of PD treatment including levodopa pharmacokinetics, dopamine kinetics, basal ganglia neurotransmission and motor action,” J Pharmacokinet Pharmacodyn, vol. 48, pp. 133–148, 2021. doi: 10.1007/s10928-020-09723-y

[37] Viceconti, Marco and Pappalardo, Francesco and Rodriguez, Blanca and Horner, Marc and Bischoff, Jeff and Musuamba Tshinanu, Flora “In silico trials: Verification, validation and uncertainty quantification of predictive models used in the regulatory evaluation of biomedical products,” Methods, vol. 185, pp. 120–127, 2021. doi: 10.1016/j.ymeth.2020.01.011

[38] Viceconti, Marco and Lanubile, Filippo and others “Position Paper: Extending Credibility Assessment of In Silico Medicine Predictors to Machine Learning Predictors,” IEEE J Biomed Health Inform, vol. 29, no. 7, pp. 5284–5290, 2025. doi: 10.1109/JBHI.2025.3552320

[39] Wakasugi, Noritaka and Takano, Harumasa and Abe, Mitsunari and others “Harmonizing multisite data with the ComBat method for enhanced PD diagnosis via DAT-SPECT,” Front Neurol, vol. 15, pp. 1306546, 2024. doi: 10.3389/fneur.2024.1306546

[40] Yngman, Gunnar and Bjugård Nyberg, Henrik and Nyberg, Joakim and Jonsson, E. Niclas and Karlsson, Mats O. "An introduction to the full random effects model," CPT:PSP, vol. 11, no. 2, pp. 149–160, 2022. doi: 10.1002/psp4.12741